Adding, Subtracting, and Multiplying Polynomials

Name _____

Algebra 2 Lesson 3-2 (fast track) Date _____ Period _____

Learning goal: I can add, subtract, and multiply polynomials and write the answer in standard form.

Language goal: I can explain each step using the words like terms, coefficient, Distributive Property, degree, and standard form.

Spanish cognates: *Propiedad Distributiva, Propiedad Conmutativa, Propiedad Asociativa.*

On the quiz: (1) Add two polynomials in a and b; write the sum in standard form (Example 1). (2) Multiply a binomial by a trinomial (Example 2A). (3) Closure of monomials (full Savvas quiz only). (4) Bonus: volume of an open box, x times length times width, multiplied out (Example 2B and the Practice). (5) Compare two volume rates (full Savvas quiz only).

Example 1: Add and Subtract Polynomials

The idea: use the Commutative and Associative Properties to group **like terms** (same variables, same exponents). Then combine like terms: only the **coefficient** changes.

A. $(6x^3 + 4x + x^2 - 7) + (2x^3 - 8x^2 + 3)$

1. Group like terms: $(6x^3 + 2x^3) + 4x + (x^2 - 8x^2) + (-7 + 3)$
2. Combine: $8x^3 + 4x - 7x^2 - 4$
3. Standard form (highest degree first): $\boxed{8x^3 - 7x^2 + 4x - 4}$

B. $(3x^2y^2 + 2xy^2 + 6x^2) - (2x^2y^2 + 3xy^2 - 2x^2)$

1. Distribute the factor of -1 to *every* term: $3x^2y^2 + 2xy^2 + 6x^2 - 2x^2y^2 - 3xy^2 + 2x^2$
2. Group like terms: $(3x^2y^2 - 2x^2y^2) + (2xy^2 - 3xy^2) + (6x^2 + 2x^2)$
3. Combine: $\boxed{x^2y^2 - xy^2 + 8x^2}$

Degree of a term = sum of its exponents: x^2y^2 (4), xy^2 (3), $8x^2$ (2), so B is in standard form.

Say it: In Part B, I _____ the -1 to every term of the second polynomial. Then I combine _____, and only the _____ changes.

Try It 1: Add or subtract the polynomials as indicated

a. $(4a^4 - 6a^3 - 3a^2 + a + 1) \quad + (5a^3 + 7a^2 + 2a - 2)$

b. $(2a^2b^2 + 3ab^2 - 5a^2b) \quad - (3a^2b^2 - 9a^2b + 7ab^2)$

c. $(3.4m^3n^2 + 5.6m^2n) - (0.7m^3n^2 - 4.1m^2n - n) + \left(\frac{6}{5}m^3n^2 - \frac{9}{4}m^2n + \frac{3}{8}m\right)$

Example 2: Multiply Polynomials

The idea: use the **Distributive Property**: multiply *every* term of the first polynomial by *every* term of the second. Then group and combine like terms.

A. $(2m+5)(3m^2-4m+2)$

1. Distribute: $2m(3m^2-4m+2)+5(3m^2-4m+2)$
2. Multiply: $6m^3-8m^2+4m+15m^2-20m+10$
3. Group: $6m^3+(-8m^2+15m^2)+(4m-20m)+10$
4. Combine: $\boxed{6m^3+7m^2-16m+10}$

Check with a box: $2\times 3=6$ products.

	$3m^2$	$-4m$	$+2$
$2m$	$6m^3$	$-8m^2$	$4m$
$+5$	$15m^2$	$-20m$	10

Like terms sit on the diagonals:
$-8m^2+15m^2=7m^2$ and $4m-20m=-16m$.

B. $(mn+1)(mn-2)(mn+4)$ Three factors: multiply **two binomials first**, then multiply that product by the third.

1. First two: $(mn+1)(mn-2)=m^2n^2-2mn+mn-2=m^2n^2-mn-2$
2. Distribute the third: $(m^2n^2-mn-2)(mn+4)=m^2n^2(mn+4)+(-mn)(mn+4)+(-2)(mn+4)$
3. Multiply: $m^3n^3+4m^2n^2-m^2n^2-4mn-2mn-8$
4. Group and combine: $m^3n^3+(4m^2n^2-m^2n^2)+(-4mn-2mn)-8=\boxed{m^3n^3+3m^2n^2-6mn-8}$

Say it: To multiply, I use the _____ to multiply every term of the first polynomial by every term of the second. Then I combine _____ and write the answer in _____.

Try It 2: Multiply the polynomials

a. $\left(6n^2-\frac{1}{2}\right)\left(n^2+n+\frac{3}{5}\right)$

b. $(mn+1)(m^2n-1)(mn^2+2)$

Practice: An open tray from sheet metal

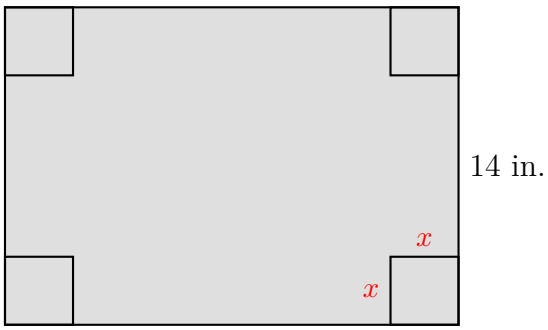
A foundry manufactures aluminum trays from pieces of sheet metal as shown. Squares of side x are cut from each corner and the sides are folded up. Let x represent the side length of each square.

a. Write expressions for the length, width, and height of the metal tray.

Length = _____

Width = _____

Height = _____



b. Write and simplify a polynomial function V to represent the volume of the tray.

Volume = length × width × height. Multiply two factors first, as in Example 2B.

$V(x) =$ _____ · _____ · _____

$V(x) =$ _____ x^3 − _____ x^2 + _____ x

IXL: Practice for the quiz

Do the starred skills first. **SmartScore 70 or above on each starred (★) skill replaces your quiz score.**

Code	Skill	Feeds
9A3 ★	Add and subtract polynomials	Quiz 1 (add, standard form)
8GN ★	Multiply polynomials	Quiz 2; Quiz 4 bonus (box)
DYB	Polynomial vocabulary	Warm-up: like terms, degree

Exit question: Practice 13 (Reason)

Polynomial function R is the difference of two degree-two polynomial functions. What are the possible degrees for R? Explain.
Hint: subtract two degree-two polynomials whose x^2 terms have different coefficients, then two whose x^2 terms have the same coefficient.

The degree of R can be _____. When I subtract, I combine _____. If the x^2 terms have the same _____, they cancel, so the _____ of R goes down (when $R \neq 0$; if every term cancels, $R = 0$ has no degree).

Before the quiz, finish these sentences.

Like terms have the same variables and the same _____.

To subtract a polynomial, I distribute -1 to _____ of its terms.

To multiply polynomials, I use the _____ and then combine _____.

A polynomial is in standard form when its terms go from highest to lowest _____.